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Subject : Management Information System - II

Assignment : 05.

1.] Define customer relationship Management (CRM) and explain its main objective.

=> Customer Relationship Management is a business Strategy and technology system used to manage a company's interactions with current and potential customer. It involves collecting customer data, analyzing it, and using it to improve relationships, enhance satisfaction, and increase profitability.

- Main Objective of CRM :

i.] Improve customer satisfaction :

- CRM helps business understand customer needs, preferences and expectations.
- It enables personalized services and quick problem resolution.
- customer feels valued, leading to higher, satisfaction level.

ii.] Enhance customer Retention :

- CRM focuses on maintaining long-term relationship with customers.
- It track customer history, purchases and interactions.
- Reduces customer churn (loss of customers).

iii) Increase Sales and Revenue .

- CRM identifies potential leads and sales opportunities .
- Helps in tracking the entire sale cycle (lead → prospect → customer)
- Improves conversion rates and overall revenue.

iv) Centralized customer data management :

- CRM stores all customer information in one place
- Includes contact details, purchase history, preferences, and feedback.
- Reduces duplication and error in data.

v) Improve customer communication :

- CRM enables communication through multiple channels : Email, phone, social media, chat.
- Ensure timely and consistent communication.
- Builds trust and transparency with customer.

2.] Explain concept of Enterprise Resource Planning (ERP) and illustrate how ERP helps modern organizations in managing their business processes ?

2.) Enterprise resource planning (ERP) is an integrated software system used by organization to manage and automate core business process in a centralized manner. it connects different department such as Finance, human resources, production, supply chain, sales and inventory into a single system.

- ERP in modern organization :

i) Improved business process integration :

- ERP connects all departments (finance, HR etc.)
- Information flow smoothly across the organization.
- eg. Sales data automatically updates inventory

ii) Better decision making :

- Provides realtime reports and analytics.
- Managers can make data-driven decisions.
- Reduces dependency of guesswork.

iii) Increased efficiency and productivity :

- Automation reduces repetitive manual tasks.
- Employee can focus on important details.
- Save times and operational costs.

iv) Enhanced Data Accuracy and consistency :

- Centralized system reduces duplication of data.
- Ensure all data department work with the same information.
- It uses minimizes error in manual work.

v) Better Financial Management :

- Track income, expenses, and financial transactions.
- Generates financial reports like balance sheets and profit & loss statement.
- Help in budgeting and financial planning.

3] Analyze the recent trends in Decision Support System (DSS) and explain their importance in business decision making in simple words

→ Recent trends in Decision Support System:

i) Use of Artificial Intelligence & Machine learning:

- Modern DSS now use AI and ML to analyze large amount of data.
- They can predict future outcomes and suggest actions.

ii) Real-time data Analysis:

- DSS now works with live data instead of old data.
- business can make quick decisions instantly.

iii) cloud based DSS:

- DSS system are now available of cloud.
- Easy to access from anywhere.

iv) Big data integration:

- DSS can now handle huge volume of data.
- Help in deeper and more accurate analysis.

v) Automation and AI agents:

- DSS is now automated with integrated tools.
- Some system can make actions automatically.

- Importance of DSS in business decision making

i) Better and faster decisions:

- DSS provides accurate data and analysis.
- Help manager make quick and correct decision.

ii) Reduce Risk:

- Predict future problem and opportunities
- Help avoid losses and make safe decisions

iii) Improves Efficiency:

- Automates data Analysis and reporting.
- Save times and efforts

4] Evaluate the role of an Expert system in organizations & explain its major applications.

> - Role of Expert System in organizations:

i) Decision support:

- Helps managers make informed decisions.
- provides solution based on stored expert knowledge.

ii) Improves Efficiency:

- Automate problem solving processes.
- Reduces time required for analysis and decision making.

ii) Reduce dependence on Human Experts :

- Organizations don't always need a human expert.
- Expert knowledge is stored and reused.
- Major Application of Expert Systems :

i) Health care / Medical Diagnosis :

- Diagnoses deases based on symptoms :
- Suggest treatments and medications.

ii) Finance and banking :

- Credit loan approval and evaluation.
- Fraud detection system.

iii) Manufacturing :

- Fault detection in machines :
- Quality control and maintainence decisions.

iv) customer Support :

- chatbots and support virtual assistance.
- Answer customer queries automatically.

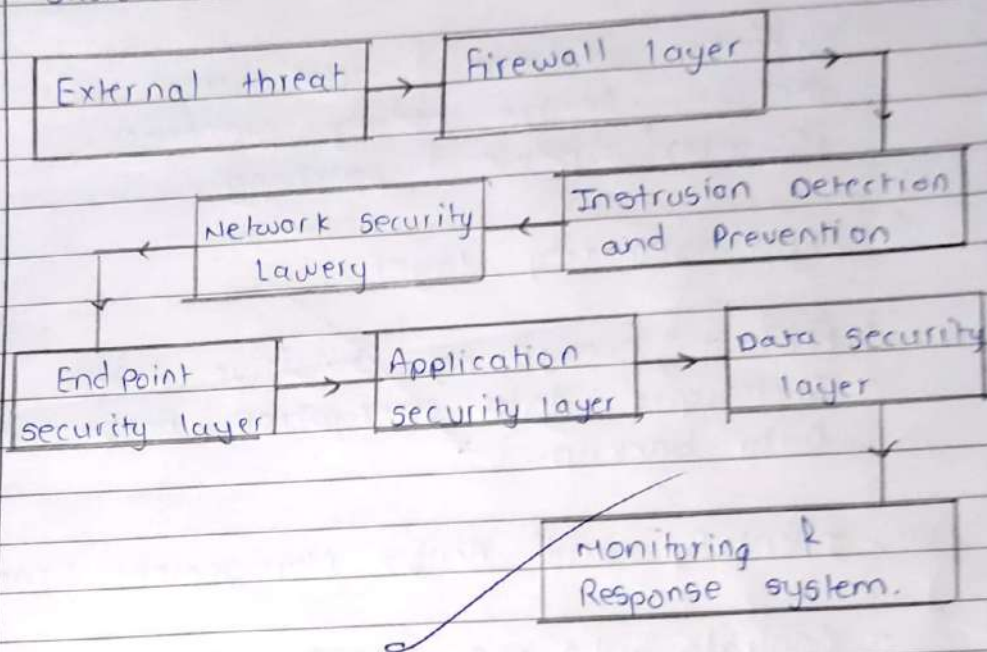
v) Human Resource Management :

- Helps in recruitment and candidate selection
- Evaluates employee performance.

6] Design a basic framework or model for inter networked security defense for protecting organizational network and explain its components

2) Inter-networked security defenses refer to a layered and interconnected security system used to protect an organisation's network from an internal & External threats.

- Basic framework :



i) Firewall layer :

- Act as an the first line of Defense.
- control incoming & outgoing network traffic.
- Block unauthorized access.

ii) Intrusion Detection & Prevention.

- IDS -> Detects suspicious activities.
- IPS -> detects & also block threats

iii) Network security layer :

- Protect internal network infrastructure.
- Includes : VPN and network segmentation.

iv) Endpoint security layer :

- Protect devices like : computer, laptop
- Tools used : Antivirus, Anti-malware.

v) Application security layer :

- Secures software & application
- includes secure coding practices, regular update & patching.

vi) Data security layer :

- Protect everything sensitive data.
- Technique data encryption, Access control, Data backup.

vii) Identity and Access Management (IAM) :

- Controls who can access what.
- Includes : user authentication, multi-factor authentication.

~~06/04/26~~